MIL-H-80003B
28 March 1974
SUPERSEDING
MIL-H-80003A
23 September 1969

MILITARY SPECIFICATION

HONING MACHINES, VERTICAL, SINGLE SPINDLE, 1-1/2 to 12 INCH BORE, 12 to 80 INCH STROKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical, single spindle, honing machines having a floating type honing tool which self-aligns with an existing bore in the workpiece. The honing machines covered herein are suitable for internal honing of open, blind-end, and interrupted bores which have been generated by a previous forming or machining operation.

* 1.2 Classification. This specification covers honing machines of the following sizes. The maximum honing diameter and stroke length required for specific procurements shall be as specified (see 6.2.1).

Maximum Honing Diameter, Inches	Stroke Length, Inches
1-1/2	12
2	12 or 25
4	12, 25, 40, or 50
6	18, 25, 35, 45, or 60
8	25, 35, 40, 45, 50, 60, or 70
10	25, 35, 40, 45, 50, 60, or 70
12	25, 40, 65, or 80

2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 3419

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SPECIFICATION

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments.	Occupational Safety and Health Administration Standards

NATIONAL BUREAU OF STANDARDS (NBS) HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS

ANS B5.1	T-Slots, Their Bolts, Nuts, Tongues, and Cutters

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

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JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

3. REQUIREMENTS

3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of internal honing operations in accordance with the requirements herein. The machine shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

* 3.2.1 Controls. All operating controls shall be located convenient to the operator at his normal workstation. All operating controls such as those used for regulating spindle rotation, dwell time, stone expansion rate, cycle time, and spark-out time, shall have a graduated dial or plate, conforming to 3.3.7, indicating the speed, time, or rate to which the control is adjusted. In lieu of the above, the machine may have devices, such as tachometers or other indicators, that indicate the actual operating speed, time, or rate at which the controlled components are operating.

* 3.2.2 Safety measures. Covers or other safety devices shall be provided for all parts of the machine presenting a safety hazard which can be guarded without substantial interference to operation of the machine. All guards shall be removable to provide easy access to the guarded parts.

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Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.2 and 6.5).

3.2.3 Lubrication. Means shall be provided to insure adequate lubrication to all moving parts. Recirculating systems, when provided, shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification of part or machine.

* 3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repair. The structure shall be capable of withstanding all forces encountered during the operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with pastes shall be used for reclaiming any defective parts.

3.2.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

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3.3.6 Threads. All threaded parts shall be in the inch system and in accordance with NBS H28.

* 3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background and securely affixed to the machine.

3.3.8 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

* 3.4 Components. The machine shall consist essentially of a base and column assembly having a single column casting or heavy steel fabrication, a spindle head, spindle, spindle rotating mechanism, spindle reciprocating mechanism, worktable, honing tool, and their related lubrication, hydraulic, coolant, and control systems.

* 3.4.1 Base and column assembly. The base and column assembly shall be either an integral casting or fabrication or a base casting or fabrication and a single column casting or fabricated structure rigidly and accurately bolted together. If separate structures, each shall have accurately machined pads at their connecting points and the structures shall be doweled and rigidly bolted together. The base and column shall be rigid, box type, structures that are internally ribbed and braced to minimize distortion and deflection and enable the machine to perform at its maximum designed honing rates while maintaining the alignment tolerances and honing accuracies specified herein. The column shall have hardened and ground ways or guide bars for supporting and guiding the spindle reciprocating mechanism. The ways, guide bars, or parts sliding thereon shall have wipers that effectively prevent entrance of coolant, abrasives, or other extraneous materials between the mating parts. All hydraulic piping, tubing, and electrical cables serving the spindle head shall either be contained within the column structure, or shall be neatly routed and securely clamped to the outside of the column. Clamping of these components shall be to the extent necessary to present a workmanlike appearance, prevent damage to the component and injury to operating personnel, but shall not restrict motion of the spindle head. Covered access openings of ample quantity and size shall be provided in the base and column structures to facilitate servicing of parts contained therein. Means shall be provided on the base for accurately leveling and securely fastening the machine to a foundation.

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* 3.4.2 Spindle head. The spindle head shall contain the spindle, spindle bearings, and the spindle rotating drive mechanism. All rotating parts within the spindle head shall be mounted on anti-friction bearings of a type, size and precision class suitable for the purpose. All moving parts shall be lubricated by an automatic and positive lubrication system, such as a submerged oil bath, a pressurized lubrication system, or an oil spray system. The lubrication system shall require no operator attention during machine starts, stops, or during honing operations. The spindle head housing shall be sealed to prevent leakage of lubricant or hydraulic fluid, if applicable, and to prevent entrance of coolant and abrasive materials.

* 3.4.3 Spindle. The spindle shall be mounted on ball, roller, or hydrostatic bearings or a combination thereof. The bearings shall be sufficient in size and of the quality necessary to rigidly support the spindle while the machine is being operated at its maximum rated capacity. The spindle shall be rotated and reciprocated by the drive mechanisms described in 3.4.4 and 3.4.5. The "axis of spindle to column face" dimension shall conform to the requirements of Table I. When specified (see 6.2.1), the above dimension, on machines having a 12 inch maximum honing diameter, shall conform to the "optional" dimension shown in Table I. The spindle runout, end play and squareness with the table work surface shall conform to the requirements of Table II (see 3.7).

* 3.4.4 Spindle rotating drive mechanism. The spindle shall be rotated by a drive mechanism consisting of a single speed electric motor, a variable speed electric or hydraulic motor, a reduction gear system, a belt drive system, or any combination of these components. The drive system shall provide either infinitely variable spindle speeds, or the number of gear selected spindle speeds, and the spindle speed range specified in Table I. Infinitely variable speed changes may be accomplished either through the use of an infinitely variable speed electric or hydraulic motor, or an infinitely variable speed belt drive mechanism. Geared speed changes may be accomplished through the use of a multi-speed geared transmission or through the use of manually changed pick-off gears. Multi-speed transmissions shall be fully enclosed and sealed to prevent leakage of lubricants or entrance of coolants or abrasives and shall have a self-contained lubrication system with means for filling, draining, and observing the lubricant level. If speed changes are by manually changed pick-off gears, it shall be possible to change from any of the available spindle speeds within the range specified in Table I to any other available speed by changing not more than one pair of gears.

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* 3.4.5 Spindle reciprocating mechanism. The spindle reciprocating mechanism shall consist of a hydraulic cylinder which reciprocates either the entire spindle head or a spindle quill located therein. The spindle reciprocating speed shall be infinitely variable over the speed range specified in Table I. The spindle reciprocating speed control shall be completely independent of the spindle rotating speed control. The reciprocating mechanism shall smoothly reciprocate the spindle without bind, vibration, or chatter, and shall smoothly accelerate, decelerate, and reverse the direction of spindle travel without shock or lost motion. Unless otherwise specified (see 6.2.1) machines in which the entire spindle head is reciprocated shall be furnished with an air operated counterbalance system or other means to insure uniform reciprocation in each direction. The machine shall be equipped with an electrically operated upper limit stop to stop the reciprocation at a predetermined position or the extreme top of the stroke after completion of the honing cycle.

* 3.4.5.1 Reciprocating reversal trips. The spindle reciprocating mechanism shall be equipped with adjustable reversal trips capable of reversing the reciprocating motion of the spindle at any desired location within the stroke range of the machine. With the machine running at its highest reciprocating speed, and while operating at a constant honing load, the location of the spindle stroke reversal points shall be repeatable to an accuracy of +0.030 inch measured at any position of the trips and at both ends of the stroke. Machines having a maximum honing diameter of 4 inches or more shall have provisions for short-stroking the reciprocating motion of the spindle. Short-stroking shall be accomplished by manually operating a short-stroke control mounted on the operator's push-button control panel. It shall be possible to short-stroke the spindle at any desired position, within the stroke range of the machine without disturbing the setting of the reversal trips, by manually operating the short-stroke control. When manual operation of the short-stroke control is discontinued, the reversal trips shall immediately resume their functions. When specified (see 6.2.1) machines having a maximum honing diameter of less than 4 inches shall have a short-stroking mechanism conforming to the above requirements.

* 3.4.5.2 Reciprocating dwell mechanism. The machine shall have an adjustable, automatically operated dwell mechanism that, when activated, will cause the reciprocation of the spindle to dwell, or momentarily stop, at the lower end of the stroke for a predetermined, adjustable time period to facilitate accurate sizing of blind or interrupted bores. The time

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period of the spindle dwell shall be infinitely adjustable from zero to not less than 1 second. When the dwell control is turned to the "off" position, the normal reciprocation cycle shall resume. The control for the dwell mechanism shall be located convenient to the operator.

* 3.4.6 Worktable. The machine shall have either a cast or fabricated, fixed position, non-adjustable type worktable. The table working surface shall be square with the axis of the spindle in accordance with the requirements of Table II (see 3.7). The table work surface shall have not less than three T-slots conforming to the requirements of ANS B5.1, and the table dimensions shall conform to the size requirements of Table I. The worktable shall have a coolant trough around its entire perimeter for collecting spent coolant with means for returning the coolant to the coolant system for reuse. When specified (see 6.2.1), the machine shall have the type worktable specified and fully described in lieu of the fixed position worktable required above.

* 3.4.7 Honing tools. The honing tools shall be of the free floating, self-aligning type utilizing universal or ball joints which allow the honing tool to follow the existing bore of the workpiece. The honing tools shall utilize a positive expansion and contracting system for expanding and contracting replaceable honing stones. The honing tools shall be compatible with the stone expansion mechanism specified in 3.4.8. Honing tools, stones, and other tooling shall be furnished as specified (see 3.9.1 and 6.2.1).

* 3.4.8 Stone expansion mechanism. The machine shall have an automatic expansion mechanism which shall rapidly expand the honing stones until they either contact the wall of the workpiece or approach the wall within a preselected distance, then expand the stones at a slower controlled rate until the desired bore size is reached, and then rapidly collapse the stones to prevent further cutting or marring of the workpiece bore while the honing tool is withdrawn. The stone expansion mechanism shall have provisions for manually adjusting, from the operator's control panel, the desired stone expansion rate. The stone expansion system shall be of the type in which the stones are expanded by a positive action mechanism which expands the stones at a continuous, uniform and preselected rate. To prevent stone breakage, the expansion system may have a mechanism that will temporarily prevent the stones from expanding at the preselected rate when tight spots are encountered in the bore, but the system shall not allow "self expansion and contraction" as the honing tool is stroked through a tapered bore. The stone expansion mechanism shall have a stone wear compensating device

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that will automatically adjust the stonefeed mechanism for each honing cycle to compensate for the amount of abrasive worn away during the previous honing cycle.

* 3.4.8.1 Spark-out (clean-up) control. The machine shall have a control that, when actuated, will stop expansion of the honing stones, while rotation and reciprocation is continued, to allow the stones to spark-out or clean up the bore. The control shall allow the time of spark-out or the number of spark-out strokes to be preselected. The control shall be mounted on the operator's control panel.

* 3.4.8.2 Stone wear indicator. The machine shall have a stone wear indicator that will warn the operator when the stones are worn to the extent that workpiece damage is imminent. The indicator shall warn the operator by either a light, a buzzer or a bell.

* 3.4.9 Guide bushing bracket. When specified (see 6.2.1), the machine shall have a guide bushing bracket that will accommodate bushings of proper size for honing bores of the sizes specified in Table I. (The bushings will be fabricated by the machine purchaser as needed, and are not required to be furnished with the machine.) The guide bracket and bushing shall assist in guiding the honing tool into and out of the workpiece bore and shall not interfere with the self-aligning action of the honing tool after it enters the workpiece bore. Guide brackets on machines with 40 inches or greater spindle travel shall incorporate a power operated raising and lowering mechanism.

* 3.4.10 Hydraulic system. The hydraulic system shall be complete and shall include all pumps, gauges, valves, piping, manifolds, cylinders, and pressure controls. Over-pressure protection shall be provided on the high pressure line from the hydraulic pump and on the low side of each pressure reducing valve to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level and shall have a capacity conforming to the requirements of Table I. The hydraulic system shall be designed to operate, under all operating conditions, at a safe operating temperature that is compatible with the oxidation and flash point characteristics of the fluid recommended for use in the system. A temperature indicator shall be provided for monitoring the hydraulic fluid temperature.

* 3.4.11 Coolant system. The machine shall have a coolant system consisting of a coolant reservoir, a motor driven coolant pump, a swarf removal system, and necessary controls, piping and tubing for directing

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the coolant to the workpiece and for returning the spent coolant through the coolant circuit, back to the coolant reservoir. The coolant reservoir shall be a separate structure attached to the base of the machine in a location that will allow easy access to the reservoir interior for cleaning and servicing. Covered access opening of ample quantity and size shall be provided to facilitate cleaning the interior of the reservoir. The coolant reservoir capacity shall conform to the requirements of Table I. The coolant pump shall be a type suitable for handling common honing coolants containing contaminants normal to honing applications. The swarf removal system shall conform to the requirements of 3.4.11.1 and shall be located in the coolant flow circuit so as to provide clean coolant to the pump suction line. The coolant pump and swarf removal system electrical controls shall allow the swarf removal system to be operated either while the honing machine is in operation or not in operation.

* 3.4.11.1 Swarf removal system. The coolant system shall have provisions for utilizing one of the external swarf removal systems conforming to 3.4.11.1.1 thru 3.4.11.1.3. The system shall remove abraded metal, loose abrasives, abrasive binder material, and other foreign materials from the coolant. (Procuring agencies are referred to 6.7 for additional information concerning selection of a swarf removal system.)

* 3.4.11.1.1 Combination magnetic separator-fabric filter. Unless otherwise specified (see 6.2.1), a combination magnetic separator-fabric filter unit shall be furnished with each machine. The unit shall be completely automatic and shall require no operator attention. The first stage of the unit shall consist of a motor driven magnetic drum, partially submerged in a coolant trough, that revolves and magnetically attracts and removes ferrous materials from the coolant. The unit shall have a roller that squeezes excess coolant from the magnetic swarf to prevent loss of coolant. The unit shall also have a device that scrapes the magnetic swarf from the revolving drum and discharges it into a disposal trough. The second stage of the unit shall consist of a roll of filter fabric arranged so that the partially cleaned coolant discharged from the magnetic separator flows through the filter fabric, which shall remove the nonmagnetic particles from the coolant. The fabric filter stage shall have a mechanism that automatically advances the roll of filter fabric, exposing a clean filter area, when the filter fabric becomes clogged to the extent that filter capacity and efficiency are impaired. The mechanism shall automatically discharge the dirty fabric, with its collected swarf, into the same disposal trough that receives the swarf from the magnetic separator, so that swarf and consumed fabric filter may be easily disposed of. Unless otherwise specified (see 6.2.1), the

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unit shall have a capacity conforming to the requirements of Table I, and shall be capable of removing all particles larger than 10 microns in size.

* 3.4.11.1.2 Magnetic separator. When specified (see 6.2.1), a magnetic separator conforming to the applicable requirements of 3.4.11.1.1 shall be furnished in lieu of the combination magnetic separator-fabric filter unit required in 3.4.11.1.1. The gallons per minute capacity shall be as specified (see 6.2.1 and 6.7).

* 3.4.11.1.3 Fabric filter system. When specified (see 6.2.1), a fabric filter system conforming to the applicable requirements of 3.4.11.1.1 shall be furnished in lieu of the combination magnetic separator-fabric filter unit required in 3.4.11.1.1. The gallons per minute capacity and filtration particle size shall be as specified (see 6.2.1 and 6.7).

* 3.4.11.2 Coolant refrigeration system. When specified (see 6.2.1), the coolant system shall be equipped with a self-contained, mechanical, refrigeration system designed to remove from the coolant, the heat generated while honing at high metal removal rates. The refrigeration system shall have the BTU per hour capacity specified (see 6.2.1 and 6.7).

* 3.4.12 Cycle timing system. The machine shall have a cycle timing system that can be preadjusted to automatically stop further expansion of the honing stones after the machine has made a preselected number of strokes or after a preselected period of time. The timing system shall allow the spindle rotation and reciprocation to continue, without further expansion or contraction of the stones, for a preselected time interval to permit "spark-out." Upon completion of the preselected "spark-out" period, the stones shall automatically retract, the honing tool shall automatically withdraw from the workpiece bore, and the spindle rotation and reciprocation shall stop at the top of the stroke position.

* 3.4.14 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall be in accordance with this paragraph. The system shall be in accordance with the applicable requirements of JIC Electrical Standards for General Purpose Machine Tools. Each machine shall draw all of its electrical power from a single 230/460 volts, 3 phase, 60 Hz circuit fed through a fused safety disconnect switch. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be

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suitable for connecting the size grounding conductor specified in Table 15-1 of JIC Standards for the disconnect fuse rating. The electrical control circuit conductors (excluding power circuit conductors) between components which are separable for shipment of the machine shall be connected through three, or less, polarized, locking type, separable, multiple-plug-in connectors having electrical ratings suitable for the application.

* 3.4.14.1 Motors. Motors shall have ball or roller bearings. Unless otherwise specified (see 6.2.1), the spindle rotation motor shall conform to the horsepower requirements shown as "standard" in Table I. When specified the spindle rotation motor shall be one of those shown as "optional" in Table I and shall be as specified in 6.2.1(o). The spindle rotating motor and spindle reciprocating motor shall be of the totally enclosed, fan cooled (TEFC) type.

3.5 Machine characteristics. Unless otherwise specified (see 6.2.1), the machine shall meet or exceed the applicable requirements of Table I.

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TABLE I

CHARACTERISTICS AND CAPACITIES

CHARACTERISTICS	1½ inch dia. X 12 inch stroke	2 inch dia. X 12 or 25 inch stroke	4 inch dia. X 12, 25, 40 or 50 inch stroke	6 inch dia. X 18,25,35,45 or 60 inch stroke	8 inch dia. X 25,35,40,45,50 60, or 70 inch stroke	10 inch dia. X 25,35,40,45,50, 60 or 70 inch stroke	12 inch dia. X 25,40,65 or 80 inch stroke
Maximum honing diameter, Inches	1-1/2	2	4	6	8	10	12
Spindle:							
Stroke length, inches	12	12,25	12, 25, 40, 50	18, 25, 35, 45, 60	25,35,40,45,50, 60, 70	25,35,40,45,50, 60, 70	25,40,65,80
Rotating speed range, RPM	270 to 870	270 to 870	150 to 600	75 to 350	75 to 350	75 to 350	75 to 350
Number of rotating speed selections	3 or infinitely variable	3 or infinitely variable	4 or infinitely variable	4 or infinitely variable	6 or infinitely variable	8 or infinitely variable	8 or infinitely variable
Reciprocating speed range, feet per minute	10 to 60	10 to 60	10 to 70	10 to 70	10 to 70	10 to 70	10 to 70
Number of reciprocating speed selections	← Infinitely Variable →						
Dimensions:							
Axis of spindle to column face, inches	6	6	12	14	15	15	Standard-21 Optional-15 (see 3.4.3)
End of spindle to top of fixed type table (Spindle in raised position), inches	← Two times stroke length plus 10 inches →						
Table working surface, inches	17x24	17x24	17x24	17x24	21x40	21x40	21x40
Motors:							
Spindle rotating motor, HP							
Standard	1	1	5	10	15	15	20
Optional (see 3.4.14.1 and 6.2.1 (o))	3	3	7-1/2	15 or 20	10,20, or 25	20 or 25	25
Spindle reciprocating motor, HP	3	3	5	10	15	15	20
Swarf removal system capacity, GPM	3	3	5	5	10	10	15
Liquid capacities:							
Coolant tank, gallons	30	30	35	50	50	50	50
Hydraulic tank, gallons	30	30	35	85	85	85	85

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

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3.6 Performance. The machine shall be capable of performing to the accuracies specified in 3.6.1 thru 3.6.3.

3.6.1 Honing accuracy. When performing the honing accuracy test specified in 4.5.2.1, the machine shall produce a bore having the following accuracies:

a. Out-of-round. The out-of-round of the finished bore, measured at random intervals throughout the length of the bore, shall not exceed:

0.0001 inch for bores up to and including 2.0000 inches diameter

0.0002 inch for bores 2.0001 thru 4.0000 inches diameter

0.0003 inch for bores 4.0001 thru 8.0000 inches diameter

0.0005 inch for bores 8.0001 thru 12.0000 inches diameter

b. Uniform diameter throughout the length of bore. The diameter of the bore, measured at random intervals throughout its length, shall be identical within:

0.0002 inch for bores up to and including 4.0000 inches diameter

0.0004 inch for bores 4.0001 thru 10.0000 inches diameter

0.0005 inch for bores 10.0001 thru 12.0000 inches diameter

3.6.2 Smoothest surface finish. When performing the smoothest surface finish test specified in 4.5.3.3, the machine shall produce a surface finish of 5 micro-inches, or better.

3.6.3 Metal removal rate. When performing the metal removal rate test specified in 4.5.3.4, the machine shall remove metal at a rate not less than:

0.05 cubic inch/minute for bores up to and including 1.0000 inch diameter

0.20 cubic inch/minute for bores 1.0001 thru 2.0000 inches diameter

0.50 cubic inch/minute for bores 2.0001 thru 4.0000 inches diameter

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0.75 cubic inch/minute for bores 4.0001 thru 6.0000 inches diameter

1.50 cubic inches/minute for bores 6.0001 thru 8.0000 inches diameter

2.00 cubic inches/minute for bores 8.0001 thru 10.0000 inches diameter

2.50 cubic inches/minute for bores 10.0001 thru 12.0000 inches diameter

3.7 Alignment accuracy. Each machine shall conform to the alignment accuracy requirements of Table II.

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TABLE II

Alignment Accuracy

Test to be Applied	Permissible Error
Spindle:	
Runout, measured at spindle nose	0.001 inch, TIR
End play	0.0002 inch
Spindle axis square to worktable surface	0.001 inch per 12 inches
Spindle head travel square to worktable surface	0.001 inch per 12 inches
Worktable:	
Working surface flat in longitudinal and transverse direction	0.001 inch per 12 inches

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3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following standard equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. Work light.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9.1 Optional tooling. Honing tools, stone holders, honing stones, drive shafts and other tooling shall be furnished as specified and fully described (see 6.2.1).

3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

3.11 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.12 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Military or Federal Specification

If there are no points on the machine requiring lubrication, the chart or plate shall be inscribed "No lubrication necessary" in lieu of the above required information.

3.13 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate

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shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input (volts, total amps, phase, frequency)

Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date of manufacture

U S

* 3.14 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.15 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the First Article.

* 4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.3.4, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection. For procurements in which a single machine is ordered, the examination in 4.4 and tests in 4.5.1 thru 4.5.3.4 need not be duplicated if First Article inspection has been performed.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.15.

4.5 Test.

4.5.1 Operational test. The machine shall be operated at no load for not less than 60 minutes. Proper operation of all controls, motors, adjusting mechanisms and accessories shall be verified during this period.

4.5.2 Alignment accuracy test. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of Table II (see 3.7).

* 4.5.3 Performance tests. Each machine shall be subjected to the tests in 4.5.3.1 thru 4.5.3.4.

4.5.3.1 Test workpiece. The workpiece for all the honing tests shall be a section of commercial, mild steel, cold drawn, seamless tubing having an inside diameter not less than 75 percent of the machine's rated honing diameter and a length not less than 75 percent of the machine's rated stroke length. No prior preparation of the workpiece may be made to correct any inaccuracies which may be inherent to the workpiece as received from the mill.

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* 4.5.3.2 Honing accuracy test. Each machine shall be tested as follows to demonstrate that it is capable of honing to the accuracies specified in 3.6.1. The test workpiece shall be honed to produce a cylindrical bore having a diameter 0.005 inch larger than the original diameter and a surface finish of 8 to 16 micro-inches. The test shall be performed using not more than one rough honing operation and one finishing operation, during which the type of stones, coolant, stone feed rate, rotating and reciprocating speeds and fixturing shall be at the option of the supplier. The time required to perform the honing accuracy test shall not exceed 30 minutes of actual honing time. The time required for tool changes, if performed, shall not be considered in this time period limitation. The finished bore dimensions shall conform to the roundness requirements of 3.6.1 (a) and the uniform diameter requirements of 3.6.1 (b).

4.5.3.3 Smoothest surface finish test. Each machine shall be tested as follows to demonstrate that it is capable of producing a surface conforming to the requirements of 3.6.2. The workpiece used for the test in 4.5.3.2 shall be finish honed at any rotating and reciprocating speed, using any type of honing stones, coolant, and at any feed rate selected by the machine supplier. The time required to perform the surface finish test shall not exceed 30 minutes of actual honing time. The surface finish of the bore shall conform to the requirements of 3.6.2.

* 4.5.3.4 Metal removal rate test. Each machine shall be tested as follows to demonstrate that it is capable of removing metal at the rates specified in 3.6.3. The workpiece used for the test in 4.5.3.3 shall be rough honed at any rotating and reciprocating speed, using any type honing stones, coolant and at any stone feed rate selected by the machine supplier. The machine shall enlarge the diameter of the workpiece by not less than 0.0100 inch. The metal removal rate shall conform to the requirements of 3.6.3.

* 4.5.4 Electromagnetic interference control test. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.10 in accordance with Table II of MIL-STD-461.

4.6 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

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5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2.1).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2.1).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Machines covered herein are intended for use in producing accurately sized bores with diametric and axial straightness, diametric roundness, and for producing controlled surface finishes and cross-hatch patterns. The machines are used to hone open-end, blind-end, or interrupted bores which have been generated by a previous forming or machining operation.

6.2 Ordering data.

* 6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Maximum honing diameter and stroke length required (see 1.2).

c. First Article approval, if required (see 3.1 and 6.3).

d. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.5).

e. Painting, if different (see 3.3.5).

f. When optional "spindle axis to face of column" dimension is required (see 3.4.3 and Table I).

g. When air counterbalance is not required for reciprocating head machines (see 3.4.5).

h. Short-stroking mechanism, when required on machines smaller than 4 inch maximum honing diameter (see 3.4.5.1).

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i. Type worktable required, if different; fully describe type worktable required (see 3.4.6).

j. When guide bushing bracket is required (see 3.4.9).

k. Swarf removal system capacity and filtration particle size, if different (see 3.4.11.1.1 and 6.7).

l. Magnetic separator system, if required in lieu of system required in 3.4.11.1.1; also specify gallons per minute capacity required (see 3.4.11.2 and 6.7).

m. Fabric filter system, if required in lieu of system required in 3.4.11.1.1; also specify gallons per minute capacity and filtration particle size (see 3.4.11.1.3 and 6.7).

n. If coolant refrigeration system is required; specify BTU per hour required (see 3.4.11.2 and 6.7).

o. Electrical system requirements, if different (see 3.4.14).

p. When motor shown as "standard" in Table I is not required, specify one of the "optional" size motors specified in Table I (see 3.4.14.1 and Table I).

q. Machine characteristics, if different from those shown in Table I (see 3.5 and Table I).

r. Standard equipment, if different (see 3.8).

s. Optional equipment, if required, fully describe (see 3.9).

t. Optional tooling required; specify type, size, and quantity of honing tools required (whether for open-end, blind, or interrupted bores, and quantity and length of stones required). Specify quantity, type, and size of stone holders required. Specify type, length, width, grit size, and material of honing stones. Specify length and type of honing tool drive shafts required (see 3.4.7 and 3.9.1).

u. Electromagnetic interference control, if required (see 3.10).

v. Type fungus proofing, if required (see 3.11).

w. Lubrication chart, if different (see 3.12).

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x. Nameplate, if different (see 3.13).

y. First Article inspection, if different (see 4.2).

z. Quality conformance inspection, if different (see 4.3).

aa. Electromagnetic interference control test requirements, if different (see 4.5.4 and 6.4).

bb. Required level of preservation-packaging (see 5.1).

cc. Required level of packing (see 5.2).

dd. Additional marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

* 6.4 Army and Air Force procurements. When equipment purchased for the Department of the Army or the Air Force requires electromagnetic interference control, such testing should be specified to be in accordance with MIL-STD-461, Class IIB equipment, except only Test Methods CE03 and RE02 should be required.

* 6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

* 6.6 Classification data. The classification of machines covered by this revision (see 1.2), has been restructured due to discontinuation of manufacture of some old sizes and current manufacture of additional sizes.

* 6.7 Coolant swarf removal systems and refrigeration systems. The following additional information should be considered in selecting the swarf removal systems and refrigeration systems described in 3.4.11.1 thru 3.4.11.2.

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(a) Coolant swarf removal systems are required for high precision, fine surface finish honing operations. The choice of the system and capacities thereof are dependent on the honing accuracies and finishes required by the user.

(b) If the user is currently operating honing machines, he might have a centralized swarf removal system of ample capacity to also accommodate the new machine, in which case, the requirements herein for a swarf removal system may be deleted.

(c) Magnetic separators remove magnetic particles only and do not remove loose abrasives, abrasive bonding, etc., from the coolant. They are ineffective when nonmagnetic materials are being honed.

(d) Fabric filters remove all magnetic and nonmagnetic contaminants larger than the filter medium pore size, but tend to clog relatively quickly. Filter media are consumable items and media cost might be relatively high. Fabric filter media cost might be alleviated, when honing magnetic workpieces, by employing combination magnetic separators-fabric filters. The magnetic separator utilizes no consumable materials, and removes a relatively large quantity of magnetic material, leaving fewer contaminants to clog the fabric filter medium. The additional first cost of the combination magnetic separator-fabric filter might be offset over a period of time by the saving in cost of fabric filter media.

(e) After reaching a conclusion regarding the type of swarf removal system required, procuring agencies should discuss, with honing machine manufacturers, the capabilities and filtration particle size required for their particular application and specify the required capacity and micron size required if different from those specified in 3.4.11.1.1.

(f) When honing at high metal removal rates, to close tolerances, or when honing parts having tendencies toward thermal distortion, consideration should be given to requiring the coolant refrigeration system described in 3.4.11.2. General industry recommendations are that the refrigeration system capacity, in BTU/hour, be computed on the following basis:

Minimum refrigeration capacity in BTU/hr =

$$\frac{\text{(Spindle rotating motor H.P. + Spindle reciprocating motor H.P.) X 12000}}{10}$$

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In the above equation, the motor horsepowers should be the horsepowers of the motors actually furnished on the machine, which may be more than the minimum horsepowers required in Table I.

6.8 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH
Air Force - 84
DSA - GS

User Activities:
Army - SM
Navy - MC

Preparing Activity:
DSA - IP

Project Number:
3419-0161

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Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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